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Proofs

Chapter Learning Outcomes

CLO

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- Know and understand different techniques for proofing.
- Use the (direct, indirect, contradiction, and by cases) proofing techniques to prove some conditional statements.

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Basic Definitions (Even Numbers)

An integer n is an even number, if there exists an integer k such that $n = 2k$.

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Basic Definitions (Odd Numbers)

An integer n is an odd number, if there exists an integer k such that $n = 2k+1$.

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Proof Techniques

- Direct Proof
- Indirect Proof (Proof by Contrapositive)
- Proof by Contradiction
- Proof by Cases

Direct Proof

We want to prove a statement of the form:

$$H \rightarrow C$$

1- Assume H is true.

2- Prove that C is true.

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Direct Proof - Example

Prove, if n is an odd number then n^2 is an odd number

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Direct Proof - Example

Prove, if n is an even number then $(-1)^n = 1$

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Direct Proof - Quiz

QUIZ

If we want to prove the following statement by a direct proof, which assumption should be considered as true?

If n is an even number then n^2 is an even number.

① n is even number

② n is odd number

③ n^2 is even number

④ n^2 is odd number

Proof Techniques

- Direct Proof ✓
- Indirect Proof (Proof by Contrapositive) ✓
- Proof by Contradiction
- Proof by Cases

Indirect Proof (Proof by Contrapositive)

Prove a statement of the form $H \rightarrow C$ by showing the contrapositive $\neg C \rightarrow \neg H$ is true.

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Indirect Proof (Proof by Contrapositive)

EXAMPLE

Prove, if 3n+2 is an odd then n is an odd number

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Indirect Proof (Proof by Contrapositive)

EXAMPLE Suppose $x \in Z$, prove, if $7x+9$ is even, then x is odd

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Indirect Proof (Proof by Contrapositive)

QUIZ

If we want to prove the following statement by the indirect proof, which assumption should be considered as true?

if 7x+9 is even, then x is odd

① 7x+9 is even

② 7x+9 is odd

③ x is even

④ x is odd

Indirect Proof (Proof by Contrapositive)

QUIZ

If we want to prove the following statement by the indirect proof, which assumption should be considered as true?

If n is an even number then n^2 is an even number.

① n is even number

② n is odd number

③ n^2 is even number

④ n^2 is odd number

Proof Techniques

- Direct Proof ✓
- Indirect Proof (Proof by Contrapositive) ✓
- Proof by Contradiction ✓
- Proof by Cases

Proof by Contradiction

We want to prove a statement of the form:

$$H \rightarrow C$$

1- Assume $\neg(H \rightarrow C)$ is true.

2- Show that you reach to inconsistent result.

$$\neg(H \rightarrow C) \equiv \neg(\neg H \vee C) \equiv H \wedge \neg C$$

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Proof by Contradiction - Example

Prove, if 3n+2 is an odd then n is an odd number

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Proof Techniques

- Direct Proof ✓
- Indirect Proof (Proof by Contrapositive) ✓
- Proof by Contradiction ✓
- Proof by Cases ✓

Proof by Cases

We want to prove a statement of the form:

$$(H_1 \vee H_2 \vee H_3 \vee \dots H_n) \rightarrow C$$

1- It suffices to show all the following are true:

$$\left.\begin{array}{l} H_1 \rightarrow C \\ H_2 \rightarrow C \\ \vdots \\ H_n \rightarrow C \end{array}\right] \text{Cases}$$

Proof by Cases - Example

Prove that for every integer n, n^2-n is an even integer

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Proof by Cases - Example

Case 1

Prove that if n is odd then n^2-n is an even.

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Proof by Cases - Example

Case 2

Prove that if n is even then n^2-n is an even.

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Proof by Cases – Multiple Choice Question

If x and y are integers and their sum x+y is even, then x-y is also even. What cases you should consider?

① Case 1: x+y even. Case 2: x+y odd.

② Case 1: x odd and y even. Case 2: x and y odd.

③ Case 1: $x \geq 0$ and $y < 0$. Case 2: $x < 0$ and $y \geq 0$.

④ Case 1: x and y even. Case 2: x and y odd.

Check - Chapter Learning Outcomes

CLO

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- Know and understand different techniques for proofing.
- Use the (direct, indirect, contradiction, and by cases) proofing techniques to prove some conditional statements.

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End of Chapter

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